



Area Under The Curve: RACE-01

Subjective Type Questions :

- Find the area bounded by the curve $y = x|x|$, x -axis and the ordinates $x = -\frac{1}{2}$ and $x = 1$.
- Find the area bounded by $x = 2$, $x = 5$, $y = x^2$, $y = 0$.
- Find the area enclosed between $y = \cos x$ & x -axis as x varies from $-\frac{\pi}{2}$ to $\frac{3\pi}{2}$.
- Find area enclosed between $y = \ln x$, x -axis and the two ordinates $x = \frac{1}{e^2}$ and $x = e^2$.
- Compute the larger area bounded by $y = 4 + 3x - x^2$ and the coordinates axes
- Consider two curves $C_1: y = \frac{1}{x}$ and $C_2: y = \ln x$ on the xy plane. Let D_1 denotes the region surrounded by C_1 , C_2 and the line $x = 1$ and D_2 denotes the region surrounded by C_1 , C_2 and the line $x = a$. If $D_1 = D_2$. Find the value of ' a '.
- Find the positive value of ' a ' for which the parabola $y = x^2 + 1$ bisects the area of the rectangle with vertices $(0, 0)$, $(a, 0)$, $(0, a^2 + 1)$ and $(a, a^2 + 1)$.
- Area of the closed figure bounded by the curves $y = 2 - |2 - x|$ and $y = \frac{3}{|x|}$
- Find the values of m ($m > 0$) for which the area bounded by the line $y = mx + 2$ and $x = 2y - y^2$ is, (i) $9/2$ square units & (ii) minimum. Also find the minimum area.
- If the area bounded by $y = x^2 + 2x - 3$ and the line $y = kx + 1$ is least. Find k and also the least area.

Answer Key

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| 1. $(\frac{3}{8} \text{ sq. units})$ | 2. $(\frac{117}{3} \text{ sq. units})$ | 3. (4 sq. units) |
| 4. $(2 + e^2 - \frac{3}{e^2} \text{ sq. units})$ | 5. $(\frac{56}{3} \text{ sq. units})$ | 6. (e) |
| 7. $(\sqrt{3})$ | 8. $(\frac{4 - 3\ln 3}{2})$ | |
| 9. ((i) $m = 1$, (ii) $m = \infty$; $A_{\min} = 4/3$) | 10. $(k = 2, A_{\text{least}} = \frac{32}{3})$ | |



Area Under The Curve: RACE-02

Subjective Type Questions :

- Find the area bounded by the curves $x = y^2$ and $x = 3 - 2y^2$.
- Find the area of the region bounded by the curves $x = \frac{1}{2}$, $x = 2$, $y = \log x$ and $y = 2^x$.
- Find the area of the region $\{(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$.
- A figure is bounded by the curves $y = \left| \sqrt{2} \sin \frac{\pi x}{4} \right|$, $y = 0$, $x = 2$ & $x = 4$. At what angles to the positive x -axis straight lines must be drawn through $(4, 0)$ so that these lines partition the figure into three parts of the same area.
- The line $3x + 2y = 13$ divides the area enclosed by the curve, $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.
- Find the area enclosed between $y = \sin x$; $y = \cos x$ and y -axis in the 1st quadrant.
- Find the area bounded by the inverse of bijective function $f(x) = 4x^3 + 6x$, the x -axis and the ordinates $x = 0$ & $x = 44$.
- Find area bounded by $y = f(x)$ & $y = f^{-1}(x)$, $x = 1$ & $x = e$ where $f(x) = e^x$.
- Let $f(x) = x^3 + 3x + 2$ and $g(x)$ is the inverse of it. Find the area bounded by $g(x)$, the x -axis and the ordinate at $x = -2$ and $x = 6$.
- Find area bounded by $y = f^{-1}(x)$, $x = 10$, $x = 4$ and x -axis given that area bounded by $y = f(x)$, $x = 2$, $x = 6$ and x -axis is 30 sq. units, where $f(2) = 4$ and $f(6) = 10$. (given $f(x)$ is an invertible function)

Answer Key

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| 1. (4 sq. units) | 2. $(= \frac{4 - \sqrt{2}}{\log_e 2} - \frac{5}{2} \log_e 2 + \frac{3}{2} \text{sq. units})$ |
| 3. (23/6 sq. units) | 4. $(\pi - \tan^{-1} \frac{2\sqrt{2}}{3\pi}; \pi - \tan^{-1} \frac{4\sqrt{2}}{3\pi})$ |
| 5. $(\frac{3\pi + 2}{\pi - 2})$ | 6. $(\sqrt{2} - 1)$ |
| 7. (60 sq. units) | 8. $(e^e - e - 1)$ |
| 9. (9/2) | 10. (22) |